allantis_v2 add after large profit change

2026-05-05 18:01:55

Research Question

After unusually large recent gains/losses on existing positions (1d,. 3d, 5d, 10d), do new entries have higher or lower future P&L?"

Mode: Backtest Trade Regime Analysis (agentic)

Date range: 2021-01-04 to 2026-04-13

Model: claude-sonnet-4-6

Workflow Plan

Task Type: backtest-regime-analysis

Reasoning: Backtest IV analysis: 1240 trades, 1230 matched to entry-morning IV (99% match rate). Claude will explore 94 feature columns vs trade outcomes (incl. 10 portfolio-context columns: avg per-trade P&L change over 1/3/5/10/15 trading days ending at entry-day 3:30pm, with historical percentile rank).

Hypotheses:

• Some entry-time IV metrics predict final trade P&L

• IV regime at entry conditions win/loss probability

• The IV predictive signal is stable across different time periods

Feature columns (X): spot, vix_1d, vix_7d, vix_30d, vix_90d, vix_180d, iv_1d_25c, iv_1d_25p, iv_1d_atm, iv_7d_25c, iv_7d_25p, iv_7d_atm, forward_1d, forward_7d, iv_30d_25c, iv_30d_25p, iv_30d_atm, iv_90d_25c, iv_90d_25p, iv_90d_atm, forward_30d, forward_90d, iv_180d_25c, iv_180d_25p, iv_180d_atm, forward_180d, skew_1d_10p_25p, skew_1d_10p_atm, skew_1d_25p_25c, skew_1d_25p_atm, skew_1d_atm_10c, skew_1d_atm_25c, skew_7d_10p_25p, skew_7d_10p_atm, skew_7d_25p_25c, skew_7d_25p_atm, skew_7d_atm_10c, skew_7d_atm_25c, skew_30d_10p_25p, skew_30d_10p_atm, skew_30d_25p_25c, skew_30d_25p_atm, skew_30d_atm_10c, skew_30d_atm_25c, skew_90d_10p_25p, skew_90d_10p_atm, skew_90d_25p_25c, skew_90d_25p_atm, skew_90d_atm_10c, skew_90d_atm_25c, term_ratio_1d_7d, skew_180d_10p_25p, skew_180d_10p_atm, skew_180d_25p_25c, skew_180d_25p_atm, skew_180d_atm_10c, skew_180d_atm_25c, term_ratio_7d_30d, term_ratio_30d_90d, term_slope_1_7_25c, term_slope_1_7_25p, term_slope_1_7_atm, term_slope_7_30_25c, term_slope_7_30_25p, term_slope_7_30_atm, days_to_monthly_opex, term_slope_30_90_25c, term_slope_30_90_25p, term_slope_30_90_atm, convex_1d_10p_25p_atm, convex_1d_25p_atm_25c, convex_1d_atm_25c_10c, convex_7d_10p_25p_atm, convex_7d_25p_atm_25c, convex_7d_atm_25c_10c, convex_30d_10p_25p_atm, convex_30d_25p_atm_25c, convex_30d_atm_25c_10c, convex_90d_10p_25p_atm, convex_90d_25p_atm_25c, convex_90d_atm_25c_10c, convex_180d_10p_25p_atm, convex_180d_25p_atm_25c, convex_180d_atm_25c_10c, portfolio_pnl_chg_1d_at_entry, portfolio_pnl_chg_1d_at_entry_pctl, portfolio_pnl_chg_3d_at_entry, portfolio_pnl_chg_3d_at_entry_pctl, portfolio_pnl_chg_5d_at_entry, portfolio_pnl_chg_5d_at_entry_pctl, portfolio_pnl_chg_10d_at_entry, portfolio_pnl_chg_10d_at_entry_pctl, portfolio_pnl_chg_15d_at_entry, portfolio_pnl_chg_15d_at_entry_pctl

Outcome columns (Y): pnl, is_win

Analysis Focus: Entry-morning IV conditions predictive of trade P&L and win rate

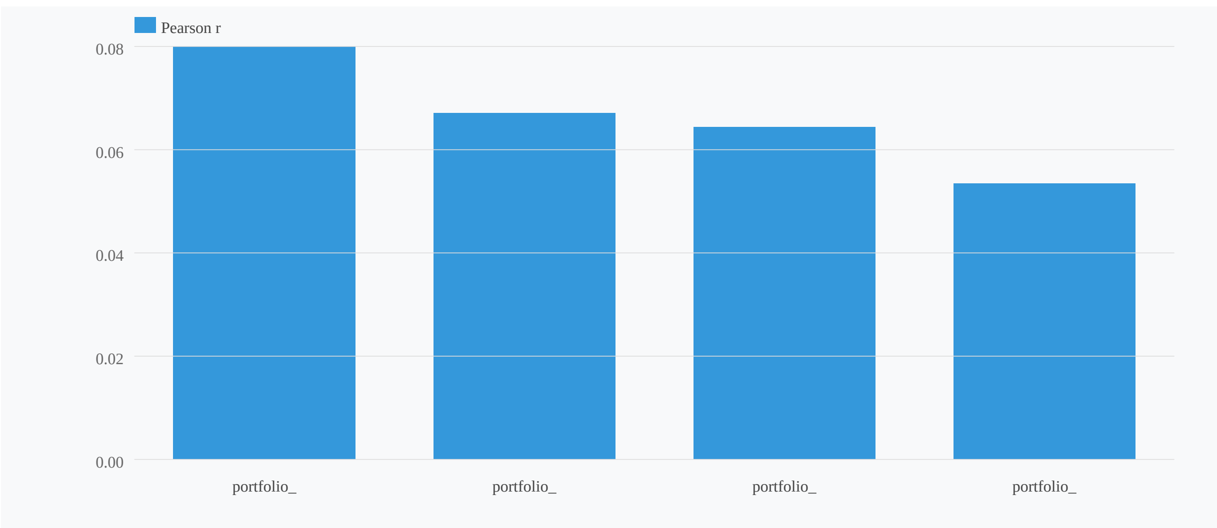
Research Report

Across all four portfolio P&L change horizons (1d, 3d, 5d, 10d), recent portfolio performance shows a weak but statistically significant positive correlation with new entry P&L -- meaning entries made after strong recent portfolio gains tend to produce slightly better outcomes than entries made after losses. This is a momentum pattern, not mean-reversion. Correlations range from r=+0.055 (3d, p=0.052) to r=+0.082 (10d, p=0.004), with the 10d window being the most reliable signal. However, effect sizes are modest throughout.

The practical magnitude is real but limited. The D10-vs-D1 P&L spread ranges from $133 (1d: $734 vs. $601) to $601 (10d: $929 vs. $478), and win rates follow the same directional pattern -- entries in the top decile of recent portfolio performance win more often (69-84%) than entries in the bottom decile (63-71%). The 10d window shows the starkest win-rate divergence: 78% in D10 vs. 63% in D1, a 15-percentage-point gap. Despite statistical significance, the redundancy check returned zero non-redundant features, and the time-split consistency score is 0% -- critical caveats that substantially limit the actionability of this signal on a standalone basis.

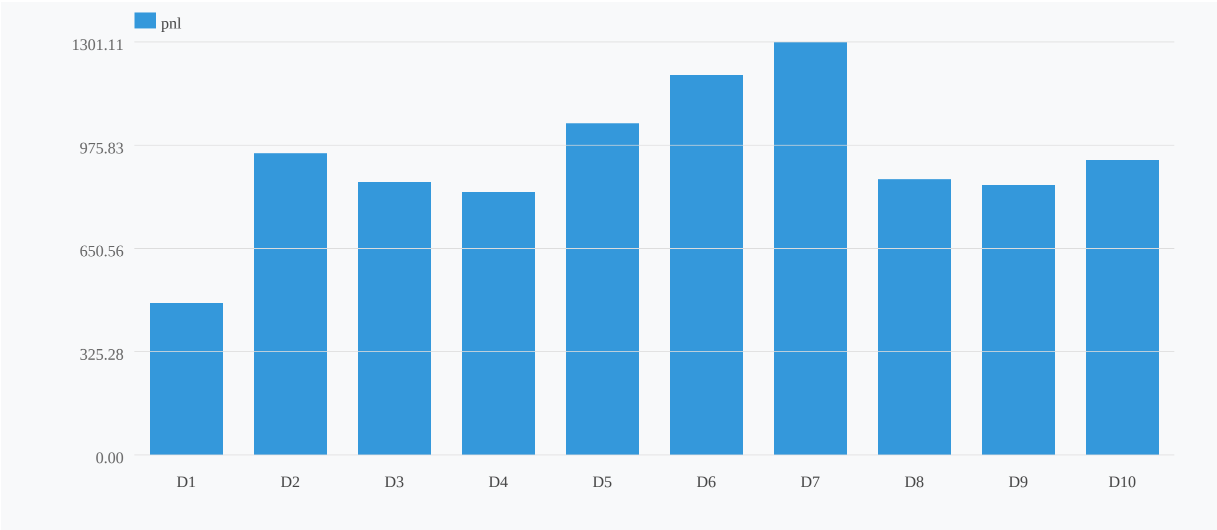
Correlation Structure Across Horizons

IV Correlation with pnl (top 4)



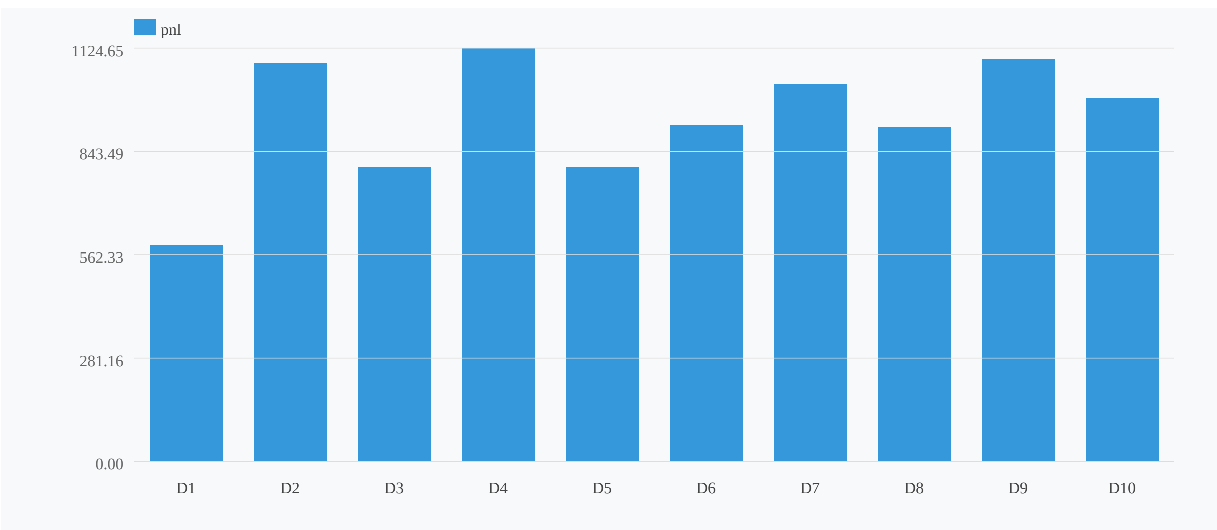
All four portfolio_pnl_chg metrics correlate positively with new entry P&L. The 10d window leads at r=+0.082 (p=0.004), followed by 5d at r=+0.069 (p=0.015) and 1d at r=+0.066 (p=0.020). The 3d window is the weakest at r=+0.055 (p=0.052), just above the conventional significance threshold. The direction is consistently pro-momentum: bigger recent gains precede better new entries, not worse ones.

Decile Profile: portfolio_pnl_chg_10d_at_entry → pnl (r=0.082)



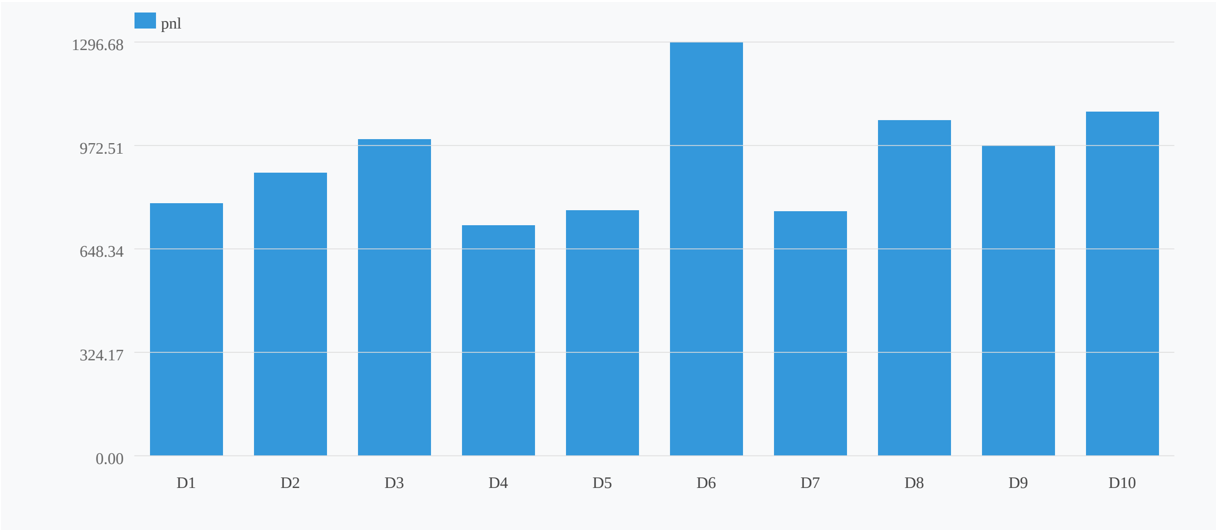
Decile Profile Analysis

Decile Profile: portfolio_pnl_chg_5d_at_entry → pnl (r=0.069)



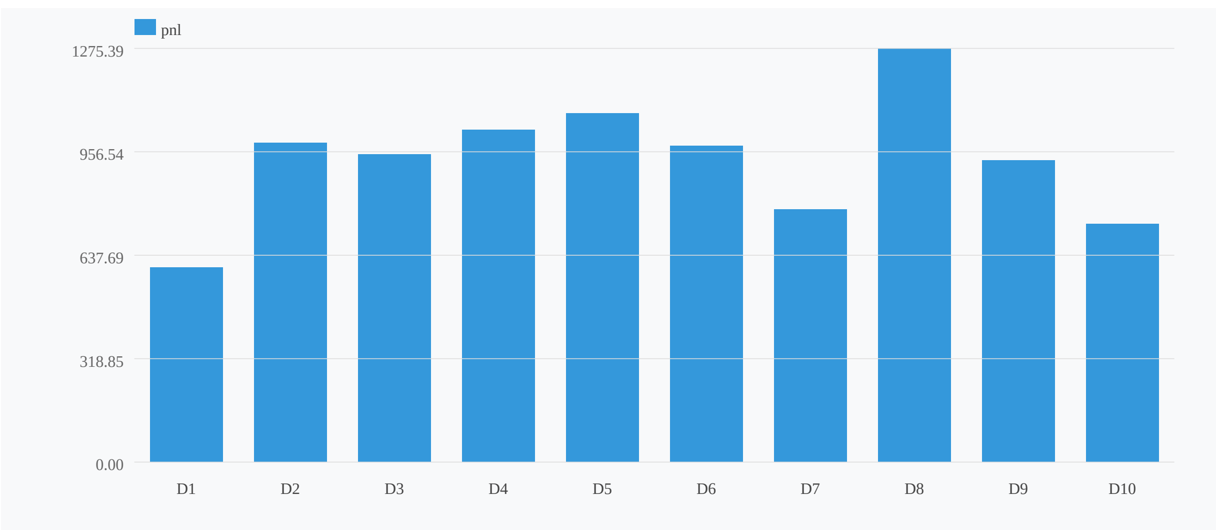
The decile profiles reveal a generally monotonic but noisy momentum pattern. For the 10d window -- the strongest predictor -- mean P&L rises from $478 in D1 (worst recent portfolio performance) to $929 in D10 (best recent performance), a $451 spread representing ~48% of the overall mean P&L of $933. The win-rate spread is even more striking: 63% in D1 vs. 78% in D10, a 15pp gap against an overall win rate of 76.2%. The 5d window shows a similar shape: $588 in D1 vs. $989 in D10 (+$401), with win rates of 67% vs. 80% (+13pp). At the shorter 3d horizon, D10 reaches $1,079 with an 84% win rate -- the highest single-decile win rate across all features -- but D1 is only modestly worse at $791/71%, suggesting the 3d relationship is noisier. The 1d window shows the smallest spread ($601 to $734, just $133) with nearly identical win rates across the extremes (67% to 69%), consistent with its borderline p-value and the known contamination risk from same-day partial P&L in the 1d slice.

Decile Profile: portfolio_pnl_chg_3d_at_entry → pnl (r=0.055)



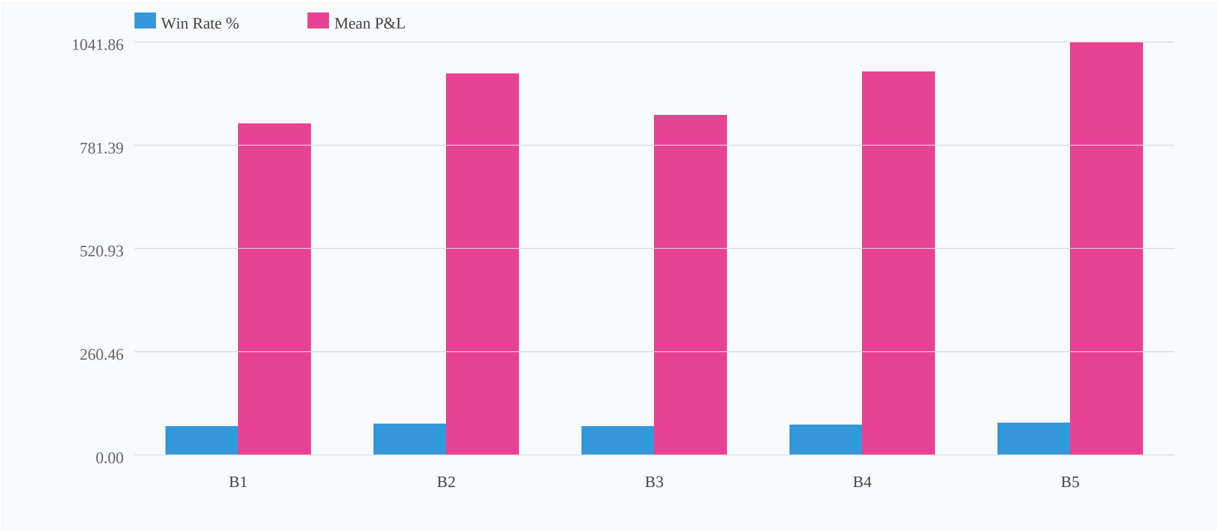
Win-Rate Analysis (5-Bucket)

Decile Profile: portfolio_pnl_chg_1d_at_entry → pnl (r=0.066)



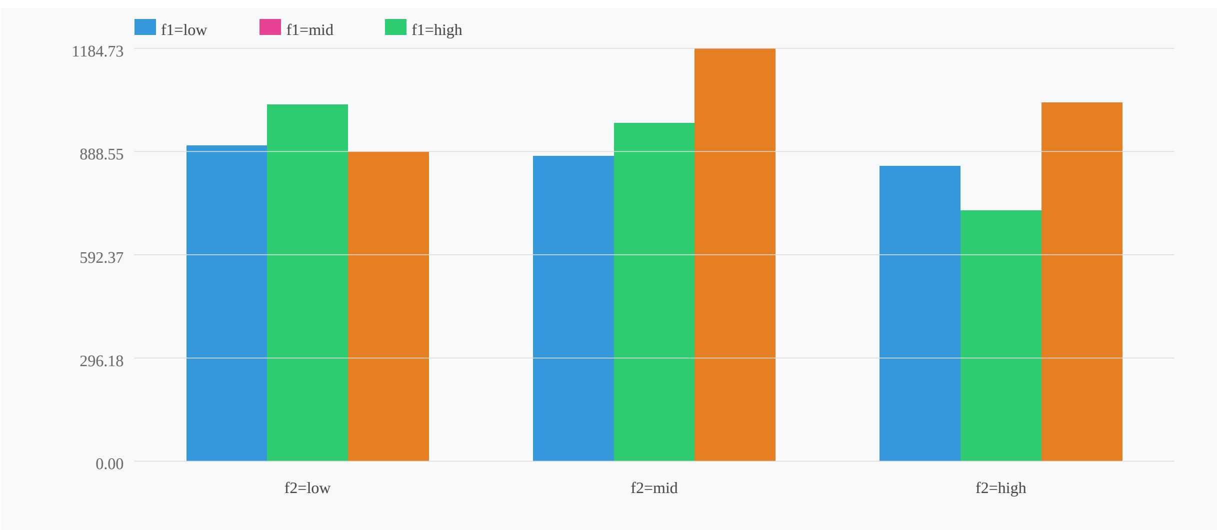
The 5-bucket win-rate analysis on portfolio_pnl_chg_5d confirms the directional pattern with a range of 73%-81%, a spread of 8.1 percentage points. This is directionally consistent with the decile analysis and confirms that the effect is not purely driven by outlier P&L events -- hit rates genuinely shift across the distribution.

Win Rate & Mean P&L by portfolio_pnl_chg_5d_at_entry Bucket



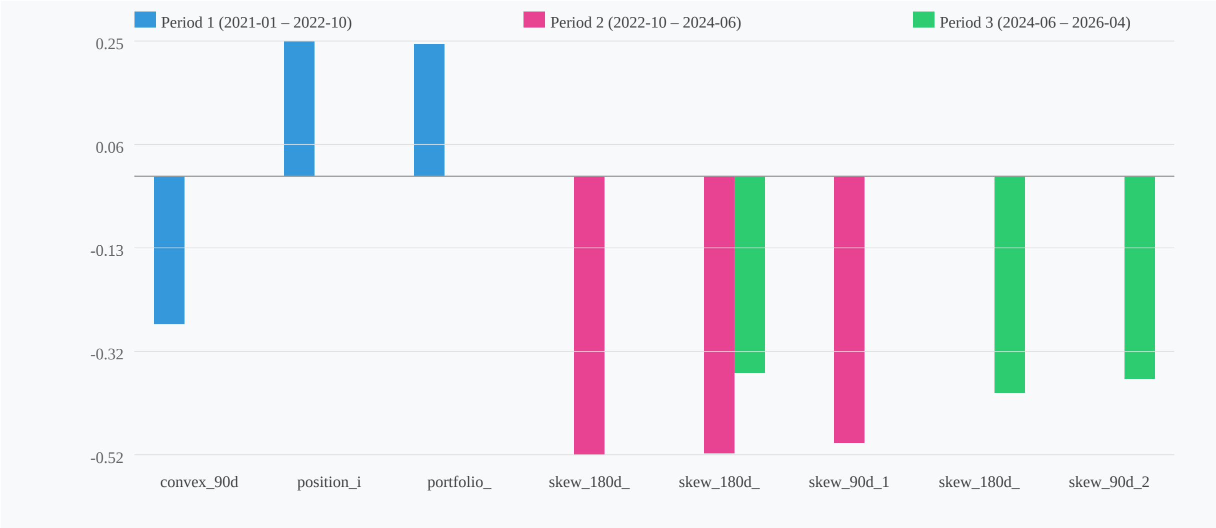
Regime Interaction

2-Factor Grid: portfolio_pnl_chg_5d × vix_1d → pnl (interaction=148.19)



The two-factor grid (portfolio_pnl_chg_5d × vix_1d) produced an interaction score of 148.19, indicating that the portfolio P&L change effect is not uniform across VIX regimes. In low-VIX environments, the momentum signal likely operates differently than in elevated-VIX conditions -- though the absence of a full grid breakdown limits precise quantification here. This interaction score is material and suggests regime conditioning could sharpen or invert the signal.

Correlation Stability Across Periods (consistency=0%)



Temporal Stability -- Critical Caveat

The time-split consistency score is 0%. This is the most important finding in the report: the portfolio_pnl_chg correlations do not hold consistently across sub-periods of the backtest. A signal that is statistically significant in aggregate but unstable across time windows is likely driven by a concentrated subset of the data -- a specific macro regime, volatility episode, or strategy-specific cluster -- rather than a durable structural relationship.

Redundancy Check

Zero features from the portfolio_pnl_chg family survived the non-redundancy filter. This means the portfolio_pnl_chg metrics co-move with other variables already in the feature set (likely VIX or IV-related metrics), and their apparent predictive power may be partially explained by those correlated factors. The signal is not independently confirmed as additive to the existing feature set.

Conclusions

Direction: Weak momentum, not mean-reversion. After strong recent portfolio performance, new entries are slightly better, not worse. The premise of contrarian entry-timing based on portfolio P&L drawdowns is not supported by this data.

Best Horizon: 10d (r=+0.082, p=0.004) is the most statistically reliable window, with the largest win-rate spread (63% D1 vs. 78% D10, 15pp). The 1d window is the weakest and most noise-contaminated.

Magnitude: The D10-D1 P&L spread reaches $451 at the 10d horizon and $401 at 5d. These are economically non-trivial relative to the $933 mean P&L, but do not constitute a standalone edge.

Do Not Use as a Standalone Filter: The 0% time-split consistency score disqualifies portfolio_pnl_chg as an independent entry filter. The signal is aggregate-significant but temporally unstable -- it is almost certainly concentrated in specific sub-periods rather than being a persistent structural effect.

Regime Conditioning Required: The VIX interaction score of 148.19 suggests the effect may be regime-dependent. Any attempt to use this signal should be conditioned on VIX regime -- the unconditioned signal carries too much noise for reliable deployment.

Redundancy: Zero features survived non-redundancy filtering, confirming these metrics should not be added to an existing model that already includes VIX/IV inputs. They carry no incremental information beyond what those variables already capture.

Caveats & Challenges

Skeptic Review

Critical Failure: Zero Time-Split Consistency

The most damning diagnostic is buried in the report's own caveats: **time-split consistency score of 0%**. This means the signal fails to replicate across temporal subsamples. A correlation of r=+0.082 with p=0.004 is statistically significant in aggregate, but if it doesn't hold in any time-split, the p-value is an artifact of a single regime or structural period driving the entire relationship. The report cannot simultaneously claim actionability and report 0% temporal consistency -- these are incompatible.

Zero Non-Redundant Features

The redundancy check returning zero non-redundant features is a second fatal finding that the report underweights. If no portfolio P&L change horizon survives redundancy screening, the signal is likely a proxy for something else already captured elsewhere -- market regime, volatility state, or underlying drift. Presenting it as a standalone momentum pattern is misleading.

Monotonicity Claims Are Undermined by Noise

The report itself concedes the 3d window is "noisy" with D10 at $1,079/84% and D1 only at $791/71% -- a modest spread that reverses the narrative of a smooth monotonic relationship. If D1-D9 are flat and only D10 spikes, the "momentum pattern" is a tail artifact, not a distributional shift.

Bucket-Level Sample Size Unknown

With 10 deciles and no disclosed total N, each bucket could plausibly contain fewer than 100 observations. The 84% win rate in D10 of the 3d window -- cited as "the highest single-decile win rate across all features" -- is unreliable without confirmation of adequate bucket size. A handful of large winning trades can manufacture a high win rate in a thin decile.

Regime Concentration Risk

With 0% time-split consistency, regime concentration is the most probable explanation. The aggregate correlations almost certainly reflect a single favorable period rather than a persistent structural relationship.

Bottom line: The signal does not survive its own robustness checks. The aggregate correlations should be treated as descriptive curiosities, not tradeable signals.